Review plan · for the teacher

Data and automation: unit check

IoT · Level 3 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Logging data you can trust
- IF/THEN: rules that act
- Power: batteries and the sun
- Networks: getting the data home

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it at school tomorrow

Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Quiz sheet · for each team

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Team _____ Date _____

1 **Before trusting a new classroom temperature sensor, what should you do first?**

- ◯ Average a week of its readings to even out errors
- ◯ Check that its chart over a day looks smooth
- ◯ Compare it with a trusted thermometer for an hour

2 **Six readings, sorted: 26, 27, 29, 30, 31 and 88 °C.**

The median is _____ °C, and the average is _____ °C.

Word bank: 38.5 · 88 · 29.5 · 29

3 **With two thresholds (fan on above 30 °C, off below 28 °C), a room at 29 °C always has the fan off.**

◯ True ◯ False

4 **A tank's level sensor may fail. Which designs fail safe?**

- ☐ Stop the pump if readings stop arriving
- ☐ Run the pump on a timer instead
- ☐ Alert the caretaker whenever the pump is stopped this way
- ☐ Keep pumping until someone sees the overflow
- ☐ Stop the pump if a reading is impossible, such as -5 %

5 **A sensor averages 0.5 mA on two AA batteries (2,500 mAh). Planning with 70 %, about how long will it last?**

◯ About 3 weeks ◯ About 7 months ◯ About 5 months

6 **For a Wi-Fi sensor that reports every minute, sending every two minutes instead does about as much for battery life as doubling the batteries.**

◯ True ◯ False

7 **The garden sensor sends over Wi-Fi once a minute, so it averages about 4 mA, day and night.**

It uses about _____ mAh a day, so a battery for 5 cloudy days needs about _____ mAh before the 70 % rule.

Word bank: 24 · 100 · 20 · 500

8 **Pupils want the school bus to report where it is, every minute, on its routes across a big city. Which network fits?**

- ◯ The school's Wi-Fi, with a stronger router
- ◯ A mobile network, with a SIM and a plan
- ◯ LoRa, with one gateway on the school roof

9 **Which of these are true of a mesh network?**

- ☐ Each extra hop costs some battery and adds delay
- ☐ Every sensor must be in range of the gateway
- ☐ More hops make the readings more accurate
- ☐ Messages hop from neighbour to neighbour to a gateway
- ☐ If one node fails, messages can often take another route

10 **Match each problem from this unit to its fix.**

85 °C from a sensor that has just powered up	• • • •	Add a second threshold
A fan flicking on and off near 30 °C	• • • •	Leave it out, and note why
Wi-Fi batteries that last 18 days	• • • •	LoRa, with a gateway
A tank 400 m away, beyond Wi-Fi	• • • •	Send less often

Answers · for the teacher

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1. **Compare it with a trusted thermometer for an hour**

   Only a reference shows whether the sensor is right.

2. **The median is 29.5 °C, and the average is 38.5 °C.**

   With six readings the median is halfway between the 3rd and 4th: (29 + 30) ÷ 2 = 29.5.

3. **False**

   Between the thresholds the fan keeps doing what it was doing.

4. **Stop the pump if readings stop arriving; Stop the pump if a reading is impossible, such as −5 %; Alert the caretaker whenever the pump is stopped this way**

   A safe system treats a missing or impossible reading as a fault, stops the pump and tells a person.

5. **About 5 months**

   70 % of 2,500 is 1,750 mAh, and 1,750 ÷ 0.5 = 3,500 hours, about 146 days: five months.

6. **True**

   Sending is almost all of the 4 mA average, so halving the sends nearly halves the current, to about 2 mA, and the life nearly doubles, at no cost, as long as data two minutes old is fresh enough.

7. **It uses about 100 mAh a day, so a battery for 5 cloudy days needs about 500 mAh before the 70 % rule.**

   4 mA × 24 hours = 96, about 100 mAh a day, and five cloudy days need 500 mAh: about 715 mAh (500 ÷ 0.7) once you plan with 70 %.

8. **A mobile network, with a SIM and a plan**

   The bus travels many kilometres across the city, so it needs a network that is everywhere it goes: mobile coverage.

9. **Messages hop from neighbour to neighbour to a gateway; If one node fails, messages can often take another route; Each extra hop costs some battery and adds delay**

   A mesh exists so that sensors out of the gateway's range can still be heard, hop by hop, and so that one broken node needn't cut anyone off.

10. **85 °C from a sensor that has just powered up — Leave it out, and note why**
    A fan flicking on and off near 30 °C — Add a second threshold
    Wi-Fi batteries that last 18 days — Send less often
    A tank 400 m away, beyond Wi-Fi — LoRa, with a gateway

    One idea per lesson: clean glitches openly, give rules two thresholds, save the battery by sending less, and choose the network from the distance and the data.